



CYART SOC INTERNSHIP

Week 2 Report : – Alert Classification and Incident Response

Name: Rucha Yeole

Organization: CyArt

Team: Security Operations Center (SOC)

Week: 2

1. Introduction

A Security Operations Center (SOC) is responsible for monitoring, detecting, analyzing, and responding to cybersecurity threats in real time. SOC analysts use structured processes and tools to identify suspicious activities and mitigate risks to organizational systems.

This task focuses on understanding three core SOC concepts: **Alert Priority Levels, Incident Classification, and Incident Response Lifecycle**. These concepts help analysts efficiently manage alerts, categorize incidents, and respond effectively to security events.

2. Objectives

The objectives of this task are:

- To understand alert prioritization based on severity levels
- To classify incidents using standard frameworks such as MITRE ATT&CK
- To study the Incident Response lifecycle based on NIST guidelines
- To perform alert triage using threat intelligence tools
- To document findings in a structured SOC format



3. Alert Priority Levels

Alert prioritization helps SOC analysts determine which alerts require immediate attention.

Severity Levels

Priority	Description	Example	CVSS Score
Critical	Immediate action required	Ransomware attack detected	9.8
High	Unauthorized admin login	Suspicious login attempt	8.5
Medium	Multiple failed login attempts	Brute force SSH	6.5
Low	Network scanning detected	Port scan activity	3.5

CVSS (Common Vulnerability Scoring System) is used to measure the severity of vulnerabilities on a scale from 0 to 10.

Higher scores indicate higher risk and urgency.

Priority	Description	Example	CVSS Score (0-10)
Critical	Immediate action required	Ransomware attack	9.8
Critical	Immediate action required	Security ancountry	9.8
Critical	Deection of potest information	Resouree comical comptaters	7.1
Critical	Describe smnis action requirists	Decand status	7.1
Critical	Grized action required	Immediate action required	6.6
Critical	Immediate action required	Ransomware attack	6.6
Critical	Immediate action required	Immediate action required	6.6
Critical	Filled action reaction	Forest unowed datation	6.6
Critical	Oavantage and intercrent action system	Virtual isus reaction	4.4
Critical	Use Immediate action required	Windows urattack attack	3.0
Critical	Oavantage of prorosed advicors	Ransomware attack	3.0
Critical	Immediate action required	Immediate action required	9.8
Critical	Consiimer auziensiled	Ransomware attack	3.0

Figure 1: SOC Alert Priority Classification Table



4. Incident Classification

Incident classification involves categorizing security events based on their nature and attack behavior.

Common types of incidents include:

- Malware
- Phishing
- DDoS (Distributed Denial of Service)
- Insider Threat
- Data Exfiltration

The **MITRE ATT&CK Framework (Adversarial Tactics, Techniques, and Common Knowledge)** is used to classify attacker techniques.

Example:

Attack Type	MITRE Technique
Phishing	T1566
Brute Force	T1110
Exploit Public Application	T1190

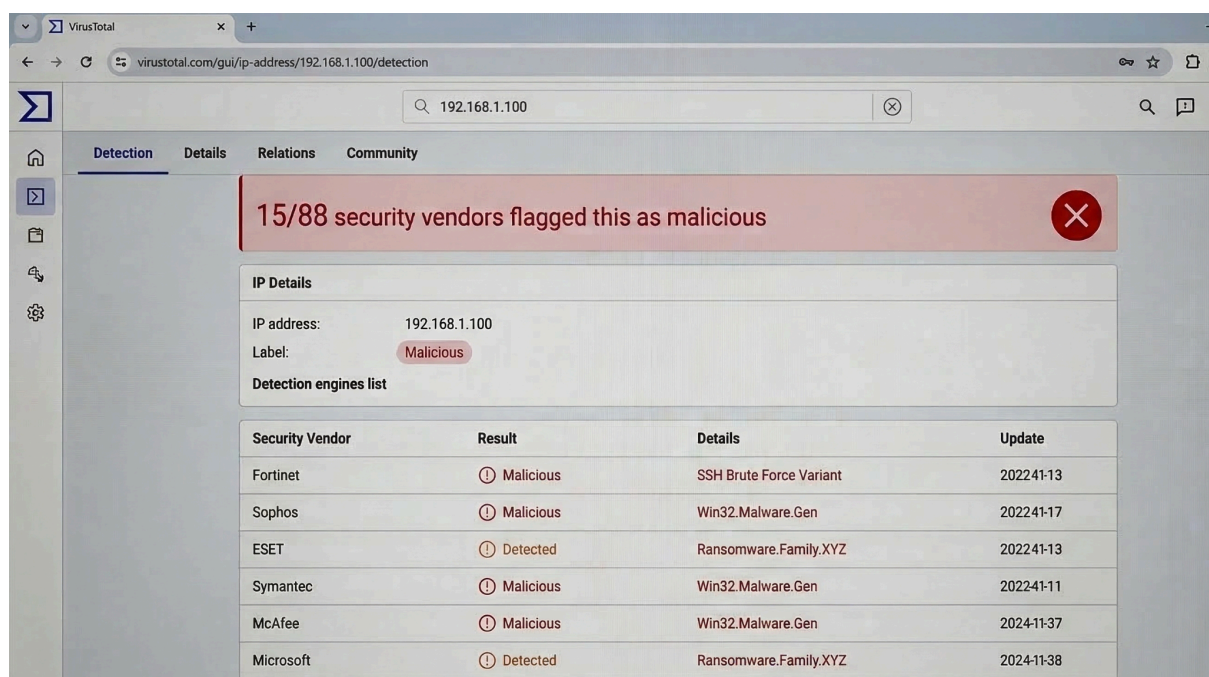


Figure 2: MITRE ATT&CK Technique T1566 (Phishing)



5. Incident Response Lifecycle

The Incident Response process follows the **NIST (National Institute of Standards and Technology) SP 800-61 framework**.

Phases:

1. **Preparation**
Setting up tools, policies, and response plans
 2. **Identification**
Detecting and analyzing potential incidents
 3. **Containment**
Limiting the spread of the attack
 4. **Eradication**
Removing the root cause of the incident
 5. **Recovery**
Restoring systems to normal operation
 6. **Lessons Learned**
Improving processes based on the incident
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6. Alert Triage and Threat Intelligence

Alert triage is the process of analyzing alerts to determine whether they represent real threats.

Threat intelligence platforms such as:

- VirusTotal
- AlienVault OTX

are used to validate suspicious indicators.

Example:

Alert ID	Description	Source IP	Priority	Status
002	Brute Force SSH Attempt	192.168.1.100	Medium	Investigated



The IP address was analyzed using VirusTotal, where multiple vendors flagged it as malicious.

	A	B	C	D	E	F	G
	Priority	Description	Example	CVSS Score			
1	Critical	Immediate action required	Ransomware attack detected	9.8			
2	High	Unauthorized admin login	Suspicious login attempt	8.5			
3	Medium	Multiple failed login attempts	Brute force SSH	6.5			
4	Low	Network scanning detected	Port scan activity	3.5			
5							
15							
16							
17							
18							
19							

Figure 3: VirusTotal Analysis of Suspicious IP Address

7. Evidence Preservation

Evidence preservation ensures that digital evidence remains intact for investigation.

It includes:

- Collecting memory dumps
- Recording file hashes
- Maintaining chain of custody

Example:

Item	Description	Collected By	Date	Hash
Memory Dump	Windows System Dump	SOC Analyst	18-08-2025	SHA256



8. Simulated Incident Scenario

A simulated security scenario was analyzed to understand the alert-to-response workflow.

An attack involving unauthorized access attempts was detected. The system generated alerts, which were analyzed and classified using SOC procedures.

The suspicious IP was validated using threat intelligence tools, and necessary actions such as monitoring and containment were considered.

This exercise demonstrated how SOC analysts detect, analyze, and respond to potential threats.

9. Conclusion

This task provided practical understanding of SOC operations, including alert prioritization, incident classification, and response procedures.

Using frameworks such as CVSS, MITRE ATT&CK, and NIST helped in structuring the analysis and improving decision-making.

The exercise highlighted the importance of efficient alert triage, threat validation, and proper documentation in handling cybersecurity incidents.

10. Tools Used

- Google Sheets
- MITRE ATT&CK Framework
- VirusTotal
- AlienVault OTX
- Google Docs / Microsoft Word